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manufacturability. "Uncooled detectors come mainly under LWIR range mentioned in the image which shows electromagnetic spectrum. Uncooled detectors are low-cost as compared to cooled and can be used for many routine applications," adds Singh.

"Cooled detectors were earlier used in thermal imaging cameras and are still being used in specific cameras designed for research and development applications. However, the thermal imagers used for industrial maintenance now use an uncooled detector," adds Thakkar.

What to consider when making a purchase

Thermography makes debugging easier and minimises the time to market. It is to be noted that the infrared imaging is done based on electrical simulation and power dissipation of the components in the board. So components that are electrical insulators cannot be evaluated by this method.

Main points to be considered are reliability of the product, after sales support and long-term warranty on the detector.

Criteria for selecting the right thermal imager

Field of view (FOV). It is the largest area that can be viewed on a thermal imager at a set distance. It is generally measured as horizontal angular degrees by vertical angular degrees.

Sensitivity. Sensitivity is basically the smallest temperature difference that can be detected. Sensitivity of a thermal imager is most often measured in terms of noise equivalent temperature difference (NETD). It is a measure of the noise in a system and expressed as degrees Kelvin (°K). The smaller the NETD number, the better the image quality and ease of identification.

Temperature range. It is the minimum and maximum temperature that this camera can measure. A thermal imager should be selected in a way that its given temperature range covers all possible temperatures on the objects you are likely to deal with.

Resolution. Detector resolution and display resolution are different. Detector resolution is an important factor that decides the quality of the image and measured data. A camera that has higher resolution exhibits better accuracy and helps measure sharper, finer details at longer distances.

Accuracy. Accuracy of measurement is the maximum deviation of the reading from the original value that the device may exhibit. It is usually expressed in percentage (%) of the scale or degrees Celsius (°C) or degrees Fahrenheit (°F). It is preferred to have an accuracy of ± 2 per cent.

Spectral response. The infrared radiations can be divided into different bands depending on their wavelength. The thermal imagers are manufactured with one of these wavelengths: short-wave or longwave. The shortwave (or near infrared) has a spectral range of 2µm to 5.6µm. The longwave (or far infrared) range is between 8µm and 15µm.

Power supply. For certain applications you can plug in the device to the mains directly or connect it to the power supply module provided.

Mount, size and weight. Some devices are hand-held and certain others are to be fixed on a surface or put on stand. The size and weight of camera is to be considered if it is to be used frequently and for longer periods.

Other features

It is important to have a suitable software provided along with the device. Certain models of IR cameras even provide built-in added features like visible-light camera, LED lamps, laser pointers, Wi-Fi, Bluetooth, etc.

Future possibilities

Innovations in technology and introduction of light-weight, highly durable materials may aid the fabrication of compact devices at lower cost. These will further increase the popularity and adoption of thermal imagers in a wide range of applications. ●

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